Assistive Keyboard for Children With Ataxia

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***Abstract*—Ataxia is a neurological condition characterized by a lack of voluntary coordination of muscle movements, significantly affecting fine motor skills, making everyday tasks such as writing challenging [1]. This paper introduces an innovative split assistive keyboard designed specifically for children with ataxia to improve their typing skills and overall computer interaction. The keyboard features an ergonomic split design, allowing for custom positioning that aligns with the user's natural hand separation and movement. The keys are also larger than those on standard keyboards, reducing the precision required for keystrokes.**

***Keywords*—ataxia; assistive technology; split keyboard design; ergonomic keyboard; children with developmental disabilities; digital inclusion;**

Fig. 1. Split Assistive Keyboard.

I. Introduction

In the field of assistive technology, the quest to create inclusive tools that meet the diverse needs of users with disabilities has led to the development of innovative solutions that bridge the gap between individuals' limitations and potential. One such advance is the design of a split assistive keyboard specifically tailored for a child with ataxia–telangiectasia, a neurological disorder that significantly impairs voluntary muscle coordination [2]. Individuals with ataxia are challenged in performing tasks that require fine motor skills, such as writing, which is an essential skill in today's digital-centric world. Recognizing the unique difficulties these children face, the split assistive keyboard emerges as an assistive tool designed to ease the typing experience while also enabling greater comfort and efficiency.

This assistive keyboard is distinguished by a split design, allowing each half to be positioned according to the natural placement of the user's palms and hands, reducing the coordination effort normally required by standard keyboards [3]. This ergonomic thinking is crucial for children with ataxia, as it accommodates their limited control over muscle movements, allowing them to more effectively operate computers and digital interfaces. This keyboard is precisely crafted to meet the specific needs of its users. Offering a mix of customization, ease of use, and assistive technology, the split assistive keyboard is a testament to the power of assistive devices in transforming the lives of people with disabilities, providing them with the tools they need to communicate, learn, and thrive in the digital age.

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II. Design

The design of the split assistive keyboard for children with ataxia involved a comprehensive examination of its ergonomic features, adaptability and technology integrations tailored to meet the specific needs of children who face challenges with motor coordination.

The split design is a feature that addresses the natural hand separation and positioning of users. Unlike traditional keyboards that force the user to adjust their hands to a fixed layout, a split keyboard allows for adjustment to comfortably fit the user's hand span and range of motion. This adaptability is crucial for children with ataxia, as it minimizes the effort required to touch the buttons and significantly reduces muscle strain and fatigue [3].

Button sensitivity and size are adapted to the limited control of motor functions characteristic of ataxia. For these reasons, mechanical key switches are best suited for sensitivity, ensuring that minimal force is required to register a keystroke, accommodating users with varying degrees of muscle strength and control [4]. The increased button size reduces the need for precision, making it easier for children to press the correct buttons.

The design prioritizes usability, durability and accessibility, with settings and adjustments that are easy to navigate. The aim is to encourage children with ataxia towards independence and confidence in using technology for communication and learning. From Fig. 2 it can be noted that different buttons have different visual indications. The keys are grouped in different colors according to the categories: vowels - in red, consonants - in green, punctuation marks and numbers - in yellow and function keys - in blue. Such a visual indication enables faster navigation on the keyboard, which, unlike the monotonously

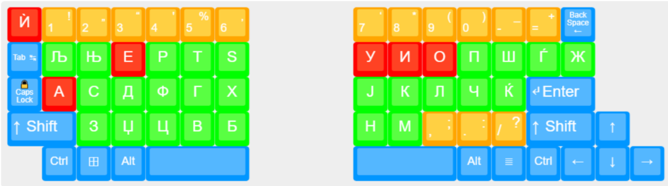

Fig. 2. Button layout and visual indicators.

colored ones, and according to our tests, reduces the search time by 1 to 2 seconds where ataxia is present.

III. Making of the Case

The durability and long-term structural integrity of a split assistive keyboard designed for a child with ataxia is an important feature. The unique demands placed on assistive devices—especially those intended for children with motor control challenges—require careful consideration of materials, design principles, and manufacturing techniques. For these reasons, the most appropriate choice for making a keyboard case is with a 3D printing machine, which is suitable for low-cost prototyping and low-volume production [5].

The choice of material significantly affects the durability and safety of the keyboard. For children with ataxia, materials must be non-toxic, hypoallergenic and robust. Thermoplastic polymers such as ABS (acrylonitrile butadiene styrene) and PLA (polylactic acid) are common choices for 3D printing due to their structural strength and ease of printing [6]. In our case, PLA polymer proved to have satisfactory characteristics. For parts that require more flexibility or impact resistance, TPU (Thermoplastic Polyurethane) can be used, especially in areas that are subject to frequent movement or pressure [7].

When designing a keyboard, the physical stress that will face during use must be considered. This includes taking into account the force of pressing the buttons, the possibility of drops or shocks and the stress on the connectors. Rounded corners can reduce stress concentrations that lead to cracks during a fall, and also reduce the risk of injury due to sharp edges.

In 3D printing, the orientation of a print substrate part and the settings used can affect its mechanical properties. Sections are generally stronger in the X and Y directions (along the substrate plane) than in the Z (height) direction [8]. Thus, critical components that will bear greater load or stress should be oriented to maximize strength in the required directions. Optimizing print settings to ensure good layer adhesion further improves the durability of printed parts.

Considering the individualized needs of children with ataxia, parts of the keyboard can be adjusted for ergonomics and ease of use. These parts were designed with modularity in mind, ensuring that they can be replaced without compromising the overall structural integrity of the device. Also, the height, width and weight of the same should be observed, in order to obtain the necessary ergonomics. This approach not only extends the life of the keyboard, but also allows for changes depending on the needs of the user over time.

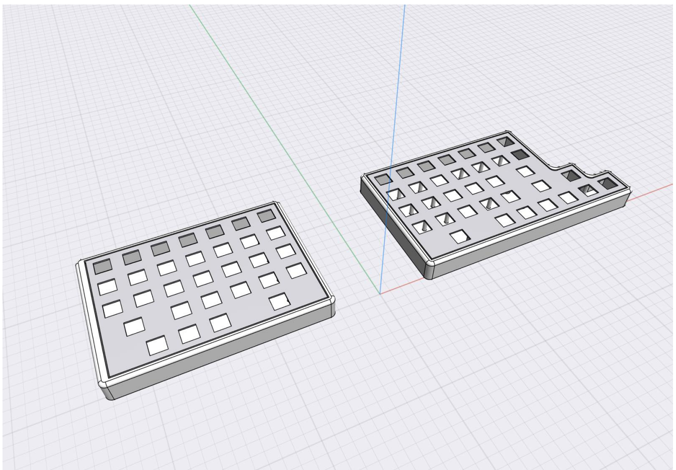

Fig. 3. Case design.

The design of the keyboard should facilitate easy assembly and disassembly, allowing maintenance and replacement of parts. Screws are recommended as opposed to glued connections, as they allow parts to be replaced or tightened if necessary. Modular designs can also allow the keyboard to be customized or reconfigured based on the user's evolving needs. The non-slip bases prevent the two halves from moving during use, which is especially important for users with ataxia who may perform erratic movements. The lightweight and compact design improves portability, allowing the user to easily transport the keyboard between different environments such as home and school.

Hardware and Software

Creating an assistive keyboard requires careful consideration of the unique hardware and software needs that meet the specific challenges these users face. Before making these kind of keyboards, it is necessary to study the hardware and software limitations in detail before making a final decision. In our process, due to the limitations, the Arduino model Arduino Micro, which is suitable for the QMK software library, were chosen as the appropriate microcontrollers.

The hardware consists of mechanical switches, on which the buttons with a larger non-standard dimension are placed. They are connected in a matrix configuration, and the communication protocol between the two halves is serial (bit banging). In our case, the first half has 5 rows and 7 columns, and the second half has 5 rows and 9 columns. When binding keys, each key should be associated with the corresponding row and column. For example, the key "A" will correspond to matrix position *[1][1]*, i.e. row 1 and column 1.

From Fig. 5 you can see that we have added a diode to each button. The diodes are added in order to be able to press 2 or more buttons simultaneously without damaging the microcontrollers or causing a disturbance in the software.

When connecting the rows and columns to the microcontrollers, it should be noted which pins are the easiest to connect to the columns and rows, so that the wires are neatly arranged.

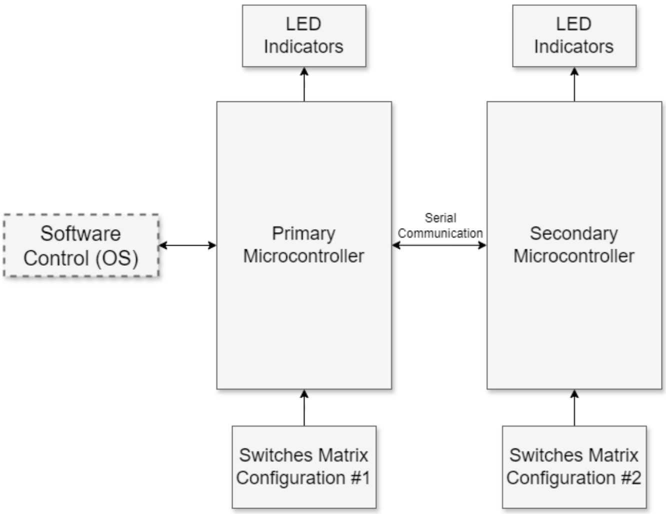

Fig. 4. Block schematic diagram.

It is also very important to note which pins are critical for serial or I2C communication (these are usually pins D0, D1, D2, D3 of the microcontroller).

The keyboard's open-source software (the QMK library) provides options for customizing and adjusting the key layout, sensitivity settings, and feedback mechanisms to suit individual preferences and needs. This level of customization ensures that the keyboard can adapt to the user's evolving skills and preferences over time.

It is important to note that the QMK library uses the microcontroller's ports, not the board's labels. For additional functionality, such as LED control, the QMK library has options that should be declared in the info.json file with the appropriate pin.

The software is designed to be compatible with various operating systems and digital platforms with the USB protocol, ensuring seamless integration with educational software, games and communication tools. This compatibility maximizes the keyboard's usefulness, making it a versatile tool for learning and everyday use.

Testing and Validation

Before finalizing the design, it is crucial to conduct rigorous testing to ensure that the keyboard meets the necessary requirements for strength, functionality and durability. This testing should simulate real-world use, including repeated keystrokes, shocks, and bending. Feedback from these tests can guide further design improvements, ensuring that the final product is both durable and functional.

For our case, we performed extensive tests on the consistency and reliability of the actuations, ensuring that minimum and variable pressure inputs are accurately registered. We subjected the keyboard to rigorous usage scenarios, strong and repeated keystrokes, to assess the durability of the keys and switches. With this we concluded that the model of the button

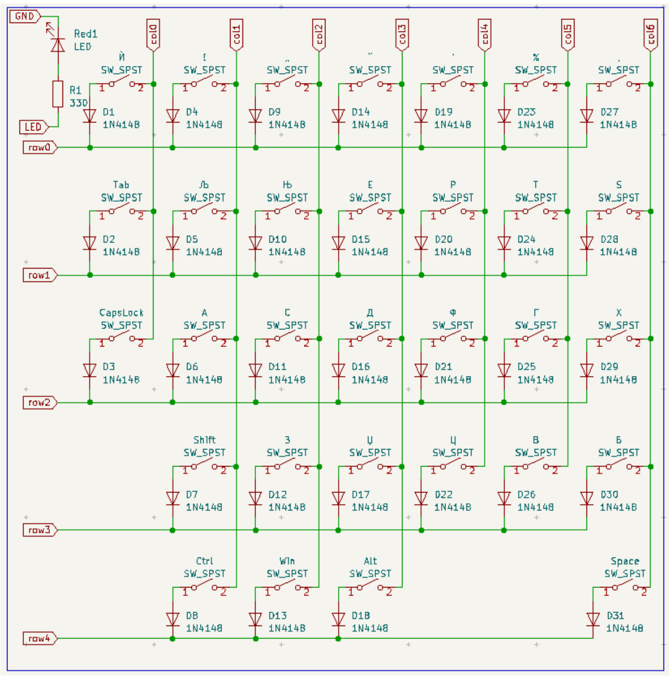

Fig. 5. Connection of mechanical switches.

holder was not suitable, so we revised and reinforced it in the second iteration of the printing process.

We've performed drop and physical stress tests to evaluate the keyboard's structural integrity, ensuring it can withstand accidental impacts. During this test, we found weaknesses in the placement and looseness of the hardware, which were adjusted and now work flawlessly.

We also tested the keyboard for long-term comfort through extended typing sessions, paying close attention to user feedback on hand and wrist strain. Apart from children with ataxia, the keyboard was also found to be comfortable for people without health disorders, where typing turned out to be more natural on a separate keyboard, without difficulty after a little training and maintaining a healthy sitting position. This allows us to evaluate the adaptability of the split design to promote a natural writing posture and reduce the effort required to write [3].

We've tested the keyboard's compatibility with different operating systems and devices to ensure broad accessibility. All relevant safety standards for electronic devices, for electrical safety and toxicity of materials, especially those intended for use by children, were observed.

Conclusion

In conclusion, the development, testing and validation of a split assistive keyboard designed for children with ataxia represents a significant step forward in creating accessible and assistive technologies. Through meticulous design considerations focused on ergonomics, customization and adaptability, this keyboard addresses the unique challenges faced by children with motor coordination difficulties, offering them a customized solution to enhance their interaction with digital environments.

An iterative development approach, driven by direct feedback from children with ataxia, caregivers and professionals, emphasizes the importance of inclusivity and accessibility in technology design. By prioritizing these values, the project contributes to breaking down barriers to digital participation for children with ataxia, giving them the opportunity to communicate, learn and engage more effectively with the world. In addition, the lessons learned and methodologies applied in this project can serve as a blueprint for future endeavors in assistive technology, highlighting the transformative impact of thoughtful user-centered design in improving the lives of people with disabilities.

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